Skinnesh: A Tool to Screen and Rank Plant Compounds for Skin-Relevant Activity and Topical Suitability

GAP2TOOL Challenge 2026

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1. Abstract

Plant compounds are commonly studied for their possible applications in skincare. Some compounds may show activity against enzymes related to pigmentation and skin ageing, such as tyrosinase, collagenase and elastase. However, experimentally testing a large number of compounds can be time-consuming and resource-intensive.

For this project, **Skinnesh**, a Python-based computational tool, was developed to screen and rank plant compounds based on their similarity to known active compounds and selected physicochemical properties related to topical suitability.

Four target channels were included: human tyrosinase, mushroom tyrosinase, collagenase/MMP-2 and elastase. Activity data obtained from ChEMBL were used to create reference datasets, which were then used for structural-similarity-based activity scoring.

A simple topical-suitability score was also included using properties such as molecular weight, logP and TPSA. The activity and topical-suitability scores were combined to generate an overall ranking.

The scoring approach was evaluated using held-out compounds from the curated ChEMBL data. ROC-AUC values of 0.865, 0.627, 0.876 and 0.943 were obtained for human tyrosinase, mushroom tyrosinase, collagenase and elastase, respectively.

As a demonstration, five plant compounds were screened using the workflow. Quercetin received the highest overall score, followed by caffeic acid, kaempferol, curcumin and resveratrol.

The main purpose of Skinnesh is to help narrow down a larger compound space and identify candidates that could be considered for further experimental investigation. The scores generated by the tool should be considered computational prioritisation signals and not proof of biological activity.

2. Introduction

Plant-derived compounds are an important area of research in skincare because many plants contain compounds with potential biological activity.

Some of these compounds may interact with enzymes involved in processes related to skin pigmentation and ageing. For example, tyrosinase is associated with melanin production, while collagenase and elastase are involved in the degradation of structural proteins.

When a large number of plant compounds are being studied, experimental testing of every compound can become expensive and time-consuming. Computational screening can be used as an initial step to reduce the number of compounds that need to be considered for experimental testing.

Based on this idea, a computational workflow was developed to take plant compounds as input, evaluate their potential activity against multiple skin-related targets, consider selected physicochemical properties, and generate a ranked list of compounds.

This workflow was named **Skinnesh**.

3. Problem Statement

Several databases and tools are available for obtaining information about compounds, biological activity and molecular properties. However, these resources are not always combined into a single workflow.

A researcher interested in screening plant compounds may need to separately:

- obtain plant compound structures,
- find known active compounds,
- calculate molecular properties,
- investigate activity against different targets,
- and compare the resulting information.

This can make the screening process more complicated, especially when a large number of compounds are involved.

A workflow that combines these steps can make the screening process more organised and reproducible.

4. Gap Identified

Existing resources related to anti-ageing compounds, including ANTIAGE-DB, were considered during the development of the project.

Several areas were identified where an additional computational workflow could be useful:

- screening multiple compounds together,
- considering multiple targets,
- keeping human and mushroom tyrosinase separate,
- including a topical-suitability component,
- providing a transparent score breakdown,
- and including computational validation.

Based on these requirements, Skinnesh was designed as an open-source command-line Python workflow.

5. Aim and Objectives

The main aim of the project was to develop a computational workflow for screening and ranking plant compounds based on their potential skin-relevant activity and topical suitability.

The objectives were:

1. To develop a Python-based workflow for screening multiple compounds.
2. To include multiple skin-relevant enzyme targets.
3. To calculate a simple topical-suitability score.
4. To compare input compounds with known active compounds.
5. To generate target-specific activity scores.
6. To combine the scores and rank the compounds.
7. To validate the scoring approach using known activity data.
8. To generate an interpretable report from the results.
9. To make the workflow reproducible and open-source.

6. Skinnesh Workflow

The main workflow of Skinnesh consists of the following stages:

Input compounds → Standardisation → Topical suitability → Activity scoring → Ranking → Validation → Report

The input compounds are first processed and their molecular properties are calculated.

A topical-suitability score is then calculated based on selected physicochemical properties.

The compounds are subsequently compared with known active compounds for the selected targets using structural similarity.

Target-specific activity scores are generated and combined with the topical-suitability score to produce the final ranking.

Validation is then performed using held-out activity data.

Finally, the results can be presented through an HTML report.

7. Target Selection

Four target channels were included in the workflow.

7.1 Human Tyrosinase

Human tyrosinase is associated with melanin production and pigmentation.

7.2 Mushroom Tyrosinase

Mushroom tyrosinase is commonly used in experimental tyrosinase studies. It was kept separate from human tyrosinase because activity against mushroom tyrosinase should not automatically be considered equivalent to activity against the human enzyme.

7.3 Collagenase/MMP-2

Collagenase-related activity was included because collagen is an important structural component of skin.

7.4 Elastase

Elastase was included because elastin degradation is associated with changes in skin structure.

The ChEMBL target IDs used were:

Target	ChEMBL ID
Human tyrosinase	CHEMBL1973

Mushroom tyrosinase	CHEMBL3318
Collagenase/MMP-2	CHEMBL333
Elastase	CHEMBL248

8. Data Sources

ChEMBL was used as the main source of biological activity data.

Activity information such as IC50 and Ki was collected for the selected targets.

The workflow was designed to support plant compound structures from public resources such as IMPPAT and PubChem.

For the final demonstration, a small example dataset containing five compounds was used:

- Quercetin
- Caffeic acid
- Kaempferol
- Curcumin
- Resveratrol

This five-compound dataset was used to demonstrate the workflow and should not be considered a complete screening of all plant compounds.

9. ChEMBL Data Curation

The raw ChEMBL activity data were processed before being used as reference data.

The workflow included the following steps:

1. Activity records were obtained for each target.
2. Relevant activity measurements were selected.
3. Records without molecular structures were removed.
4. Exact activity relations were retained where appropriate.
5. pChEMBL values were calculated.
6. Compounds were classified according to activity thresholds.

7. Reference and validation datasets were created.

Compounds with a pChEMBL value of **5.0 or higher** were treated as active, while compounds with a value of **4.0 or lower** were treated as inactive for the validation procedure.

Values between these thresholds were not treated as confident active or inactive examples.

The resulting datasets were:

Target	Raw records	Curated records	Reference actives	Validation compounds
Human tyrosinase	433	163	101	35
Mushroom tyrosinase	1,945	868	518	122
Collagenase/MMP-2	5,000	2,886	2,211	471
Elastase	5,000	2,380	1,706	631

10. Topical Suitability

A simple topical-suitability component was included in Skinnesh to provide an additional physicochemical factor during ranking.

The score uses properties such as:

- Molecular weight
- logP
- TPSA

These properties are combined to generate a topical-suitability score.

The score is intended only as a computational heuristic. It does not predict actual skin penetration, formulation performance, toxicity or clinical effectiveness.

Therefore, a high topical-suitability score indicates that a compound fits the physicochemical criteria used in the workflow, rather than proving that the compound would work effectively as a topical ingredient.

11. Activity Scoring

Known active compounds were used as reference compounds for each target.

The structures of the input compounds were compared with the reference compounds using structural similarity.

The resulting similarity information was used to generate target-specific activity scores.

Each compound therefore receives separate scores for:

- Human tyrosinase
- Mushroom tyrosinase
- Collagenase
- Elastase

These scores are then combined to obtain an activity component for the overall ranking.

The activity scores represent computational prioritisation signals. They are not equivalent to experimentally measured IC50 values, inhibition percentages or binding affinities.

12. Multi-Target Ranking

After the activity scores and topical-suitability score have been calculated, they are combined to generate an overall score.

This allows compounds to be ranked based on both their predicted activity profile and their physicochemical suitability according to the scoring system.

The individual scores are also retained in the output.

This makes it possible to understand the contribution of each target and the topical-suitability component to the final ranking.

13. Validation

Validation was included to determine whether the scoring approach showed useful ability to distinguish active and inactive compounds.

Held-out compounds from the curated ChEMBL datasets were used for this purpose.

Three main evaluation metrics were used.

ROC-AUC

ROC-AUC measures how well the scoring method can distinguish active compounds from inactive compounds.

A value close to 0.5 represents approximately random discrimination, while higher values indicate better discrimination.

PR-AUC

PR-AUC measures the relationship between precision and recall and is useful when the number of active and inactive compounds is not balanced.

EF@10%

Enrichment factor at 10% measures whether active compounds are enriched among the highest-ranked 10% of the validation compounds.

14. Validation Results

The validation results obtained were:

Target	N	Actives	Inactives	ROC-AUC	PR-AUC	EF@10%
Human tyrosinase	35	32	3	0.865	0.986	1.094
Mushroom tyrosinase	122	60	62	0.627	0.699	1.877
Collagenase/MMP-2	471	444	27	0.876	0.992	1.061
Elastase	631	591	40	0.943	0.994	1.051

The highest ROC-AUC was obtained for elastase, with a value of **0.943**.

Collagenase also showed good discrimination with an ROC-AUC of **0.876**.

Human tyrosinase showed an ROC-AUC of **0.865**.

Mushroom tyrosinase showed weaker discrimination, with an ROC-AUC of **0.627**.

However, mushroom tyrosinase showed an EF@10% of **1.877**, indicating enrichment of active compounds among the highest-ranked validation compounds.

15. Interpretation of Validation

The validation results indicate that the structural-similarity-based scoring approach provides useful ranking information for several of the selected targets.

The strongest performance was observed for elastase, followed by collagenase and human tyrosinase.

The lower ROC-AUC for mushroom tyrosinase indicates that the same approach was less effective for this target.

This target-specific variation is important and shows why validation was included in the workflow instead of assuming that the scoring method would perform equally well for every target.

The results support the use of Skinnesh as a computational prioritisation tool, but they do not establish experimental predictive accuracy.

16. Validation Limitations

The human tyrosinase validation set contained only three inactive compounds.

Because of this small number of negative examples, the human tyrosinase validation metrics should be interpreted cautiously.

The validation datasets also differed in size and composition between targets.

In addition, ChEMBL activity records can come from different experimental studies and assay conditions.

Therefore, the validation results provide an indication of the behaviour of the computational scoring method, but they should not be treated as a complete measure of its performance on new experimental data.

17. Example Plant Compound Screening

The final demonstration was performed using five plant compounds.

The ranking obtained was:

Rank	Compound	Topical Suitability	Activity Mean	Overall Score
1	Quercetin	0.930	0.698	0.790
2	Caffeic acid	1.000	0.605	0.763
3	Kaempferol	0.862	0.668	0.745
4	Curcumin	0.959	0.542	0.709
5	Resveratrol	1.000	0.374	0.624

18. Top-Ranked Compound: Quercetin

Quercetin received the highest overall score in the example dataset.

The scores obtained were:

- Topical suitability: **0.930**
- Human tyrosinase: **0.641**
- Mushroom tyrosinase: **0.512**
- Collagenase: **1.000**
- Elastase: **0.638**
- Activity mean: **0.698**
- Overall score: **0.790**

The high overall score resulted from the combination of its activity scores and topical-suitability score.

This result does not establish that quercetin is experimentally more effective than the other compounds.

It indicates that quercetin was ranked highest according to the computational criteria used by Skinnesh.

19. Biological Interpretation

The main purpose of Skinnesh is to reduce a larger compound space into a smaller group of candidates that can be considered for further experimental investigation.

For example, compounds with high activity scores across one or more targets and reasonable physicochemical properties could be prioritised for laboratory testing.

The separate target scores also provide information about why a compound received its ranking.

A compound may show a stronger computational signal for collagenase than for tyrosinase, while another compound may show a more balanced profile across multiple targets.

This provides more information than a single final score alone.

However, the computational results must be followed by experimental testing before biological activity can be confirmed.

20. Human vs Mushroom Tyrosinase

Human and mushroom tyrosinase were deliberately maintained as separate target channels.

This was done because activity against mushroom tyrosinase should not automatically be interpreted as activity against human tyrosinase.

Keeping the two targets separate allows differences in the computational activity profiles to be observed.

It also prevents the two target types from being combined into a single score without considering their biological differences.

21. Reproducibility

Skinnesh was developed as a command-line Python workflow.

The project repository contains:

- Python source code
- Example data

- Configuration files
- Dependency files
- Data preparation scripts
- Validation code
- Report-generation code
- Documentation
- Example outputs

The main commands include:

Display available commands

```
None
python -m skinnesh.cli --help
```

Rank compounds

```
None
python -m skinnesh.cli rank \
  --input data/example_plant_compounds.csv \
  --references data/reference_actives.csv \
  --output results/demo
```

Generate an HTML report

```
None
python -m skinnesh.cli report \
  --ranked results/demo/ranked_compounds.csv \
  --output results/demo/report.html
```

Run validation

```
None
python -m skinnesh.cli validate \
  --validation data/validation.csv \
  --references data/reference_actives.csv \
  --output results/validation
```

These commands allow the workflow to be reproduced without manually repeating the calculations.

22. Output

The ranking output contains information including:

- Compound name
- Canonical SMILES
- Molecular weight
- logP
- TPSA
- Topical-suitability score
- Human tyrosinase activity score
- Mushroom tyrosinase activity score
- Collagenase activity score
- Elastase activity score
- Activity mean
- Overall score

An HTML report can also be generated from the ranked results.

The report includes an interpretation warning to make it clear that the scores are computational prioritisation signals rather than proof of biological efficacy.

23. Multifunctionality

Skinnesh combines several steps into a single workflow.

These include:

1. Compound processing
2. Physicochemical property calculation
3. Topical-suitability scoring
4. Multi-target activity scoring
5. Compound ranking
6. Validation
7. Report generation

This allows a set of input compounds to be processed from initial structures to a final ranked output.

24. Limitations

Several limitations are present in the current version of Skinnesh.

Structural similarity does not guarantee biological activity

A structurally similar compound may not show the same biological activity as a known active compound.

Topical suitability is a heuristic

The topical-suitability score does not predict actual skin penetration, formulation stability, toxicity or clinical effectiveness.

Performance differs between targets

The validation results showed different levels of performance for different target channels.

ChEMBL data are not completely uniform

Activity measurements can originate from different experiments and assay conditions.

The example dataset is small

Only five compounds were used in the final demonstration. Therefore, the example ranking should not be considered a general ranking of plant compounds.

Experimental validation is required

The final candidates would need to be tested experimentally before biological or cosmetic conclusions could be made.

25. Future Work

Several improvements could be made in future versions of the tool.

A larger plant-compound dataset could be screened to provide a more extensive demonstration.

The topical-suitability component could be improved using more advanced models rather than simple physicochemical rules.

Additional skin-related targets could also be incorporated.

Machine-learning models could be compared with the current structural-similarity approach to determine whether improved prediction could be achieved.

More carefully designed inactive and decoy datasets could also improve the validation procedure.

Finally, the highest-ranked compounds could be experimentally tested to determine how well the computational ranking agrees with actual biological activity.

26. Original Contribution

The main contribution of Skinnesh is the integration of several screening steps into a single reproducible workflow.

Compound structures, physicochemical properties, biological activity information and validation were brought together into one computational pipeline.

The separation of human and mushroom tyrosinase was also maintained to avoid treating the two targets as identical.

The output provides individual target scores as well as an overall ranking, allowing the results to be interpreted rather than simply presented as a final number.

27. AI Usage Disclosure

AI tools were used during the development of the project for brainstorming, coding assistance, debugging, documentation and organisation of project materials.

28. Conclusion

Skinnesh is a Python-based computational workflow developed to screen and rank plant compounds based on their potential skin-relevant activity and topical suitability.

Four target channels were included: human tyrosinase, mushroom tyrosinase, collagenase/MMP-2 and elastase.

ChEMBL activity data were curated and used to create reference and validation datasets. The structural-similarity-based scoring approach showed useful discrimination for several of the targets, although performance varied between targets.

In the example screening of five plant compounds, quercetin received the highest overall score, followed by caffeic acid, kaempferol, curcumin and resveratrol.

The results demonstrate how computational screening can be used to prioritise compounds before experimental testing.

However, the scores generated by Skinnesh should be treated as computational prioritisation signals and not as experimental evidence of biological efficacy.

The tool provides a reproducible starting point for screening plant-derived compounds and identifying candidates for further investigation.

29. Final Summary

Skinnesh provides a workflow for:

- Screening multiple plant compounds
- Calculating basic molecular properties
- Estimating topical suitability
- Scoring compounds against four skin-relevant targets
- Separating human and mushroom tyrosinase
- Ranking compounds
- Validating the scoring approach
- Generating an HTML report
- Producing reproducible results

The overall purpose is to reduce a large compound space into a smaller set of candidates that can be considered for further experimental investigation.